PowerNile Co — Annual Business Intelligence Report 2023

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Chapter 1: Executive Summary

Key Metrics (Top 5 Figures)

- **Total Headcount:** 80 employees (Energy and Mining sectors represent the largest workforce block with 28 employees, accounting for 35% of total labor).
- **Total Direct Revenue:** 187,415,292 (distributed among Fuel Stations: 132,926,146; Solar Energy: 44,168,100; and Real Estate Sales: 10,321,046).
- **Active Projects:** 7 active projects (out of 15 total projects distributed between completed and planning stages, with a total active budget of 59,000,000).
- **Highest Revenue & Profit Sector:** Fuel Stations sector with total revenues of 132,926,146, acting as the primary and largest cash flow engine.
- **Major Financial Risk (Real Estate Concentration):** Stagnation in PowerNile Tower sales (only 2 out of 8 units sold, against blocked book value and very low return relative to expenditure).

Three Clear Strengths (Derived from Data)

- **Structural & Operational Stability:** 88.8% reliance on Full-Time contracts builds high operational stability, protecting talent and preserving institutional knowledge.
- **Financial Planning Efficiency in Completed Projects:** Successful closure of 6 completed projects with a 5.6M budget and 98.7% spend percentage, indicating high estimation accuracy and zero budget overruns.
- **Revenue Diversification with Strong Energy Client Base:** Flexible solar sales combining institutional procurement (government and corporations exceeding 35.6M) with high demand for inverters and lithium batteries.

Three Operational Risks or Gaps (Requiring Urgent Action)

- **Key-Man Risk & Project Management Bottleneck:** Operational responsibility for 3 major active strategic real estate projects concentrated solely under Dr. Wedad Hamdan, creating a critical failure point in her absence.
- **Field Wage Inequality Gap:** Sharp salary disparity in the Mining sector where top administrative positions command extremely high salaries while the field team (7 employees) starts at 5,200, threatening retention.
- **Operational & Marketing Stagnation in Real Estate:** Lowest revenue among all sectors at 10,321,046 with acute sluggishness in PowerNile Tower sales, requiring immediate repricing or marketing strategy overhaul.

Chapter 2: Human Resources Analysis

The following query measures employee count, average salary, maximum, minimum, and total payroll per sector:

```
SELECT d.division AS sector, COUNT(e.employee_id) AS headcount, ROUND(AVG(e.salary), 0) AS avg_salary, MIN(e.salary) AS min_salary, MAX(e.salary) AS max_salary, SUM(e.salary) AS total_payroll FROM employees AS e JOIN departments AS d ON e.department_id = d.department_id WHERE e.salary IS NOT NULL GROUP BY d.division ORDER BY total_payroll DESC;
```

Sector	Headcount	Avg Salary	Min Salary	Max Salary	Total Payroll
Mining	13	10192.0	5200	18000	132500
Energy	15	8767.0	6300	17000	131500
Real Estate	10	12380.0	6800	20000	123800
Fuel	13	7085.0	4800	14000	92100
Sales	8	9375.0	8000	13000	75000
IT	6	11333.0	7500	18000	68000
Finance	5	11800.0	8000	19000	59000
HR	6	8700.0	6000	14000	52200
Legal	3	14167.0	12000	17000	42500

Energy and Mining represent the largest sectors with 28 employees (35%), consuming a total payroll of 264,000. Real Estate records a high average salary of 12,380 reaching up to 20,000. Administrative sectors (IT, Finance) maintain smaller headcounts with high average salaries.

Employment Types Distribution

```
SELECT employment_type, COUNT(*) AS count, ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER (), 1) AS percentage FROM employees GROUP BY employment_type ORDER BY count DESC;
```

Employment Type	Count	Percentage
Full_Time	71	88.8
Part_Time	5	6.3
Contract	4	5.0

High reliance on Full-Time employment (88.8%) builds high organizational stability.

Chapter 3: Fuel Stations Performance

Comparing revenue, transaction counts, and fuel types between Khartoum and Omdurman stations:

```
SELECT d.department_name AS station, fs.fuel_type, COUNT(fs.sale_id) AS transactions,
ROUND(SUM(fs.total_amount), 0) AS total_revenue, ROUND(AVG(fs.total_amount), 0) AS
avg_per_transaction FROM fuel_sales AS fs JOIN departments AS d ON fs.station_id =
d.department_id GROUP BY d.department_name, fs.fuel_type ORDER BY d.department_name,
total_revenue DESC;
```

Station	Fuel Type	Transactions	Total Revenue	Avg Per Transaction
Omdurman Fuel Station	Petrol	180	281325301.0	1562918.0
Omdurman Fuel Station	Diesel	180	224038569.0	1244659.0
Omdurman Fuel Station	Gas	180	162786026.0	904367.0
Khartoum Fuel Station	Petrol	180	258223754.0	1434576.0
Khartoum Fuel Station	Diesel	180	239259106.0	1329217.0
Khartoum Fuel Station	Gas	180	163635390.0	909086.0

Both stations recorded 180 transactions per fuel type. Khartoum excelled in Diesel and Gas, while Omdurman led in Petrol. Total revenue difference between stations is only 700,000, indicating close operational efficiency.

Chapter 4: Solar Energy Sales

Top-selling solar products by quantity and revenue:

```
SELECT category, product_name, COUNT(*) AS times_sold, SUM(quantity) AS units_sold,
SUM(total_price) AS total_revenue FROM solar_sales GROUP BY category, product_name ORDER BY
total_revenue DESC;
```

Category	Product Name	Times Sold	Units Sold	Total Revenue
Inverter	10 kW Three Phase Inverter	21	233	15145000
Inverter	5 kW Hybrid Inverter	25	260	9100000
Battery	200 Ah Lithium Battery	28	308	7700000
Panel	550W Bifacial Solar Panel	21	221	3978000
Battery	AGM 150Ah Battery	24	229	3435000
Accessory	Panel Mounting Structure	32	319	2552000
Panel	400W Monocrystalline Panel	22	184	2208000
Cable	4mm Solar Cable (meter)	27	334	50100

Chapter 5: Projects & Financial Risks

Project health dashboard tracking budgets, spend percentages, and status:

```
SELECT p.project_name, d.department_name, d.division, p.status, p.budget, p.spent, p.budget - p.spent AS remaining_budget, ROUND(100.0 * p.spent / p.budget, 1) AS spent_pct, CASE WHEN ... END AS financial_health FROM projects AS p JOIN departments AS d ON p.department_id = d.department_id ORDER BY spent_pct DESC;
```

Project Name	Department	Division	Status	Budget	Spent	Spent %	Health
Solar Plant - Genad	Solar Energy Equip.	Energy	Completed	2000000	2000000	100.0	Critical
Solar Water Pumping	Solar Energy Equip.	Energy	Completed	6600000	1590000	100.0	Critical
Khartoum Fuel Station Ext.	Khartoum Station	Fuel	Completed	300000	295000	98.3	Critical
Energy Domes - Khartoum	Energy & Desal.	Energy	Completed	300000	295000	98.3	Critical
Geological Survey - Kemla	Geology & Survey	Mining	Completed	1000000	980000	90.0	Critical
Solar Marketing Campaign	Sales & Marketing	Sales	Completed	200000	195000	97.5	Critical
Nile Complex - Residential	Real Estate Dev.	Real_Estate	Active	15000000	9000000	60.0	Good

Chapter 6: Real Estate Development

Apartment sales summary and payment status by building:

```
SELECT p.project_name, COUNT(u.unit_id) AS total_units, SUM(CASE WHEN u.status = 'Sold' THEN 1 ELSE 0 END) AS sold_units, SUM(CASE WHEN u.status = 'Available' THEN 1 ELSE 0 END) AS available_units, SUM(u.list_price) AS total_list_value, SUM(u.sale_price) AS total_sales_revenue FROM projects AS p JOIN real_estate_units AS u ON p.project_id = u.project_id GROUP BY p.project_id, p.project_name ORDER BY total_sales_revenue DESC;
```

Project Name	Total Units	Sold	Available	List Value	Sales Revenue
Nile Residential Complex - Apartments	40	27	7	14179310	9292058
PowerNile Commercial Tower	20	2	6	9015390	1028988

Nile Residential Complex achieved top revenue (9.29M) with 27 units sold. PowerNile Commercial Tower faces severe stagnation with only 2 units sold out of available offerings.

Chapter 7: Human Team Performance (Solar Sales Reps)

```
SELECT e.full_name AS salesperson, COUNT(ss.sale_id) AS total_deals, SUM(ss.quantity) AS
units_sold, SUM(ss.total_price) AS total_revenue, ROUND(AVG(ss.total_price), 0) AS
avg_deal_size FROM solar_sales AS ss JOIN employees AS e ON ss.salesperson_id = e.employee_id
GROUP BY e.employee_id, e.full_name ORDER BY total_revenue DESC;
```

Salesperson	Deals	Units Sold	Total Revenue	Avg Deal Size	Rating
Dalia Farid Mekki	43	465	11481250	267006.0	Excellent
Sophia Williams	41	464	8845050	215733.0	Excellent
Grace Adeyemi	38	374	8727800	229679.0	Excellent
Hamza Elnisr Babiker	45	454	8590100	190891.0	Excellent
Ahmed Elyas	33	331	6523900	197694.0	Excellent

Chapter 8: Comparative & Ranking Analysis

PowerNile Total Valuation Report

```
WITH hr_summary AS (...), fuel_summary AS (...) SELECT * FROM hr_summary UNION ALL SELECT *
FROM fuel_summary ...;
```

Category	Metric	Value
Human Resources	Annual Payroll Total	9319200
Fuel Stations	Total Fuel Revenues	1329268146.0
Solar Energy	Total Equipment Sales	44168100
Projects	Total Project Budget	68500000
Real Estate	Total Sold Units Revenue	10321046

Chapter 9: Summary & Recommendations

- **Redesign Commercial Real Estate Pricing & Marketing Strategy:** Overhaul marketing and sales approaches for PowerNile Tower to break acute stagnation (only 2 units sold).
- **Leverage Fuel Cash Flow to Support Solar Energy:** Channel profits from the highly lucrative fuel sector (132.9M) toward expanding import lines for "Inverters," which dominate solar sales at 24.24M.
- **Diversify Solar Customer Portfolio:** Direct sales representatives to focus on private corporate clients (strong average deal size of 211,107) to reduce credit concentration and over-reliance on government procurement (21.49M).
- **Update Salary Scales & Implement Fair Salary Bands:** Close extreme wage gaps in the Mining sector between management and field workers (starting at 5,200) to protect against job leakage and turnover.
- **Decentralize Project Supervisory Responsibilities:** Relieve administrative congestion and operational overload concentrated on Dr. Wedad Hamdan by redistributing oversight across active capital projects.

PowerNile Co — Business Intelligence Division

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